

Impact of news on asset behaviour: return, volatility and liquidity in an intra-day setting

Xiang Yu, Gautam Mitra, Keming Yu

Brunel University

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Table of Contents

1	Introduction	2
1.1	Why sentiment analysis	2
1.2	Characterising Asset Behaviour	3
1.3	Role of Liquidity in Asset Behaviour.....	4
1.4	Modelling Architecture and Choice of Assets.....	5
1.5	Outline of the paper.....	5
2	Data	5
2.1	Market Data	6
2.2	Sentiment and News Meta data	6
2.3	The Chosen Data set.....	7
3	Measures: Return, Volatility, Liquidity, Sentiment and Impact	11
3.1	Return, Volatility and Liquidity Measures	11
3.2	Sentiment Measure	12
3.3	Impact Score	12
4	The Model	15
5	Computational Results and Validation	16
5.1	In Sample.....	16
5.2	Out-of-Sample	21
6	Future Research	26
7	Discussion and Conclusions	26
8	References	27
9	Appendix.....	30

Abstract

We report an empirical study of a predictive analysis model for equities; the model uses high frequency (minute-bar) market data and quantified news sentiment data. We have introduced a derived measure of news impact score which takes into consideration news flow and decay of sentiment. Given that trading strategies take into consideration three important characteristics of an asset, namely, return, volatility and liquidity, our model is designed to predict these three parameters for a collection of assets. Our study shows that inclusion of news impact data improves the prediction of volatility and liquidity. The minute-bar market data as well as intraday news sentiment metadata have been provided by Thomson Reuters.

1 Introduction

1.1 Why sentiment analysis

A major plank in the development and application of behavioural finance is the consideration of bounded rationality as introduced by Nobel laureate Herb Simon (Simon, 1964). It follows from the theme of bounded rationality and later works of behavioural theorists/economists, Kahneman and Tversky, Sheffrin, Shiller that human beings in general and retail investors in particular are influenced by various psychological imperatives of 'fear, grid and exuberance'. This is in sharp contrast with the postulates of neoclassical theories of rational behaviour and scrupulous application of logic in decision making. Thus behavioural finance in many ways determine the risk attitudes and the investment goals of the (high net worth) individuals: so called HNIs who account for the majority of the invested wealth.

The neoclassical finance theory had embraced the efficient market hypothesis (EMH) which has become the cornerstone of investment science. The doubts about EMH and recent surge of interest in behavioral finance (Kahneman and Tversky, 1979, Kahneman, 2002, Sheffrin, 2008) not only opened the debate and exposed the limits of EMH but also reinforced the important role of sentiment and investor psychology in market behaviour. The central tenet of EMH is information arrival, that is, news is rapidly digested by rational stakeholders; yet the work of Shiller (2000), Hais (2010), and Barber and Odean (2008) reinforce the irrational contrarian and herd behaviour of the investors. Today the availability of sophisticated computer systems facilitating high frequency trading (Goodhart and O'Hara, 1997) as well as access to automated analysis of news feeds (Tetlock, 2007, Mitra and Mitra, 2011) set the backdrop for computer automated trading which is enhanced by news. How investment strategies may harness sentiment of news events and also that of the market continue to be actively studied and reported by researchers in the investment community (Kahn, Dion, Brar, Hafez). Also see "Inside the Investor's Brain" (Wiley, 2007) by

Richard Peterson. In this study we investigate how predictive models of asset behaviour can be used as a precursor to developing trading strategies.

1.2 Characterising Asset Behaviour

The majority of research on asset behaviour has focused on analysing historical market data and the construction of models which fit the data and best represent the true information held in such datasets. The key features studied in these models are return and volatility of stock price, which are considered in the decision making stage of trading. In this report, however, we introduce two further (novel) concepts with a view to enhance these classical methods. Firstly, our model looks at three characteristics of an asset, namely return, volatility, and liquidity. The consideration of liquidity, the additional feature, provides knowledge on the condition of markets and more importantly indicates whether a profitable signal can be successfully executed or not. Simply being able to determine a profitable position in the markets, through observation of stock price return and volatility, is not sufficient if in fact the actual trade is not available. Secondly, the investigational data set is expanded with the incorporation of news sentiment meta-data in addition to the historical market data. The data sets are fused together to form two intra-day time series on a minute bar scale. This data fusion enables us to extract more information from the trading markets and construct a predictive analytics model for three parameters which characterise asset price behaviour.

The asset price parameters have been traditionally represented by two methods: predictive modelling and explanatory modelling. One of the most well-known explanatory models is the factor model (Fama and French, 1992). They capture return by extending the Capital Asset Pricing Model (CAPM), which was independently proposed by Treynor (1961), Sharpe (1964), Lintner (1965), and Mossin (1966), to include market capitalisation size and book-to-market ratio as explanatory factors. Such types of models can be categorised under 3 groups, that is macroeconomic, fundamental and statistical factor models depending on the choice and nature of these factors and how the respective models are calibrated. Although factor models have dominated this field of finance, a weakness is in their failure to quickly update changes in market conditions. The structure of these models is only single period inhibiting the incorporation of relevant past information at a sufficient speed. Parameters are updated through calibration but only at a slow pace where the model adapts. Mitra, Mitra and diBartolomeo (2009) have shown how the incorporation of news enhances the results of factor models and leads to an early prediction of changes in volatility. We believe that news sentiment should also be considered as an explanatory factor in the pricing of assets and therefore return. However, our approach turns to another class of models that better predict volatility in a time-varying framework – generalised autoregressive conditional heteroskedasticity (GARCH) models.

By incorporating the additional factor of news sentiment in GARCH models, the near term volatility can be estimated using past news events over many periods. This near term estimation may be obtained using a number of alternative

predictive models. Established models in this class are linear regression models (Stambaugh, 1999, Robertson and Wright, 2009), autoregressive (AR) models, moving average (MA) models and generalised autoregressive conditional heteroskedasticity (GARCH) (Bollerslev, 1986) models for volatility in particular. Our work utilises the family of AR and GARCH models within which news sentiment is introduced as an exogenous variable. Among the many applications of predictive modelling, the construction of preliminary stages to scenario generation is the most common practice. In the context of financial assets, scenario generators for the possible movements in stock price are formed from a predictive model for stock price.

Research studies which address the relationship between stock price movements and news sentiment have been growing in recent years (Tetlock, 2007, Barber and Odean 2008, Mitra, Mitra and diBartolomeo 2009, Leinweber and Sisk, 2009, Sinha, 2011). The earliest work that incorporates news articles in the influence of stock prices dates back to Niederhoffer (1971), where the font size of news print was used to determine the relative importance of news events and measure how they affected the stock market. Later, news was categorised into scheduled and non-scheduled news with researchers exploring their respective effects on stock returns and volatility (Lee 1992, Engle 1998). Tetlock (2007) was one of the first to use merely news sentiment data as a comparison to the movement in stock prices and trading activity.

1.3 Role of Liquidity in Asset Behaviour

Traditionally research on asset behaviour has focused on the study of two main features, namely, stock price and volatility of stock prices. Since the financial crisis of 2008 with the collapse of Lehman Brothers and the subsequent banking crisis and bank bail outs in the UK and other European countries the importance of liquidity has become paramount and is widely acknowledged by the finance community. Another implication of the financial crisis has been that regulators have imposed stricter liquidity requirements for banks, especially those active in the low latency trading activities (Gomber et al. 2011, AFM 2010). So for trading activities, the availability of liquidity to traders and brokers is of paramount importance. Hence many of the recent studies involving impact of news consider the implication on liquidity. For instance, Gross-Klaussman and Hautsch (2011) use the bid-ask spread, trading volume and market depth as proxies for liquidity with results showing greater increase of bid-ask spreads during news releases as opposed to market depth which does not differ much. Furthermore, Riordan, Storkenmaier et al. (2013) found that liquidity increases with news releases that have positive or neutral sentiment whereas negative news sentiment gives a corresponding decrease in liquidity.

It is well accepted in the trading community that liquidity has the role of monitoring market conditions and assessing the viability of orders decided by trading algorithms by taking into consideration the spread and the depth of the market. Given that margin requirement is an important and defining aspect of trading it can be argued that liquidity is an important determinant in deriving

trading strategies and should be introduced as a parameter in predictive models which characterise asset behaviour. There are a variety of definitions which explain the role and measure of liquidity mainly in the context of trading. Spread measures view liquidity from the point of the cost that one has to bear for immediate trade, in other words the viability of orders, and typical measures include the effective spread and the bid-ask spread. Depth measures consider liquidity as the effect of large orders on a particular asset hence looking at market conditions, and are often measured using traded volume, order volume or Kyle's λ (Kyle, 1985).

1.4 Modelling Architecture and Choice of Assets

In our empirical study we have considered 53 highly traded stocks taken from two exchanges, namely London Stock Exchange (FTSE) and New York Stock Exchange (Dow Jones); these stocks belong to nine industry sectors. Our aim is to find out whether taking into consideration news flow and news sentiment in addition to the market data leads to superior prediction of asset behaviour. We study the behaviour of equities as these are the most actively traded assets in the high frequency world and our aim is to ultimately derive high frequency trading strategies. This framework is highly applicable and relevant to the trading world as it supports decision-making models that is, strategies, which select trading portfolios with multiple assets.

1.5 Outline of the paper

The rest of the paper is organised as follows: In Section 2 we discuss the data instances that are used in our empirical study. In Section 3 we introduce the impact measure and explain how this is derived using the sentiment measure and the news flow. In section 4 we set out the predictive models which describe price, volatility and liquidity of assets. In Section 5 we discuss the findings of the study and present our conclusions. The appendix includes some further explanation of the impact measure introduced in Section 3.

2 Data

Our modelling architecture uses two streams of time series data: (i) The market data which is given at the minute bar level and includes bid price, ask price and the execution price, (ii) News meta data as supplied by Thomson Reuters. An explanation of the structure and the nature of the news meta data is given in Appendix B.

2.1 Market Data

The high frequency intraday market data is compiled on a minute-bar scale for 53 assets and cover nine sectors from banking, retail, oil to technology and communication. The data fields of the market data are set out in Table 2.1.

<i>Data Field</i>	<i>Field Name</i>	<i>Description</i>
1	#RIC	Reuters instrument code individually assigned to each company.
2	Date	In the format DD-MM-YYYY.
3	Time	In the format hh:mm:ss, given to the nearest minute.
4	GMT Offset	Difference from Greenwich Mean Time.
5	Type	Type of market data – in this case “Intraday 1 min”
6 - 9	Open; High; Low; Last	Open, high, low and last prices for the corresponding minute
10	Volume	Volume of trades in one minute
11	Ave. Price	Average price
12	VWAP	Volume weighted average price calculated for the corresponding minute
13	No. Trades	Number of trades in one minute
14 - 17	Open Bid; High Bid; Low Bid; Close Bid	Open, high, low and close bid prices for the corresponding minute
18	No. Bids	Number of bid orders placed
19 – 22	Open Ask; High Ask; Low Ask; Close Ask	Open, high, low and close ask prices for the corresponding minute
23	No. Asks	Number of ask orders placed

Table 2.1: Description of all the data fields for a company in the market data.

For those minutes that were missing in the data, the last price from the minute before would be used. Such situations occurred in the data as a reflection of no bid or ask prices during that time.

2.2 Sentiment and News Meta data

News analytics data is presented in a meta data format where news is given characteristics such as relevance, novelty and sentiment scores according to an individual asset. The analytical process of producing such scores is fully automated from collecting, extracting, aggregating, to categorising and scoring. The result is an individual score assigned to each news article for each characteristic using scales from 0-100 or probabilities. Although we use intraday news data, for a given asset the number of news stories, hence data points are

variable and do not match with the time frequencies of market data. The attributes of news stories used in our study are Relevance and Sentiment¹.

2.3 The Chosen Data set

A selection of 53 assets from FTSE100 and Dow Jones 30 across 9 different sectors is chosen for our empirical study. Table 2.2 lists these sectors and state the number of companies chosen within each sector. Summary statistics of the selected assets are given in Table 2.5. Companies with large market capitalisation were picked for their wide coverage of news; this consequently guarantees a sufficient number of data points in the time series of news meta data. In addition, the total market capitalisation of all selected assets is \$4.36trillion (US stocks: \$3.34trillion, UK stocks: \$1.01trillion). This is a good representation of the entire stock market in US and UK as a large proportion of the market has been taken into account.

<i>Sector</i>	<i>Description</i>	<i>No. of companies chosen</i>
1	Banking	8
2	Insurance	6
3	Pharmaceuticals	6
4	Oil & Gas	7
5	Manufacturing	3
6	Retail	9
7	Telecommunications	4
8	IT & Technology	6
9	Media	4
		Total: 53

Table 2.2: Break down of the chosen assets into their respective sectors.

The market data for each asset consists of open, high, low and close prices as well as bid and ask prices for each minute throughout the day from 02/01/2008 – 31/12/2008. From this, we extract the data from 08:00-18:30 each day to make up a **trading day** such that pre-trade and post-trade hours are included. The reason for this extension of hours is so that any news sentiment captured outside of trading hours can be incorporated into our predictive model to avoid any loss of information (impact of news). Close prices are used to calculate log-return and volatility.

The general trend in asset prices for 2008 can be seen from Figure 2.3; since this was a period of negative (worsening) sentiment all prices are found moving downward. Simultaneously on a secondary Y-axis, the number of news for the same period is plotted showing the impact of news on asset prices.

¹ See Appendix B for a detailed description of these attributes.

The news meta data for the chosen assets were selected under the filter of relevance score, that is, any news item that had a relevance score under the value of 70 was ignored and not included in the data set. This ensured with a high degree of certainty that the sentiment scores to be used are indeed focused on the chosen asset and is not just a mention in the news for comparison purposes for example. In Table 2.5, the number of news for each asset is considerably reduced, even halved in some cases, once the filter of relevance is applied as seen from the column of relevant news. Thus indicating that news sentiment of low relevance will definitely not be included in the chosen dataset.

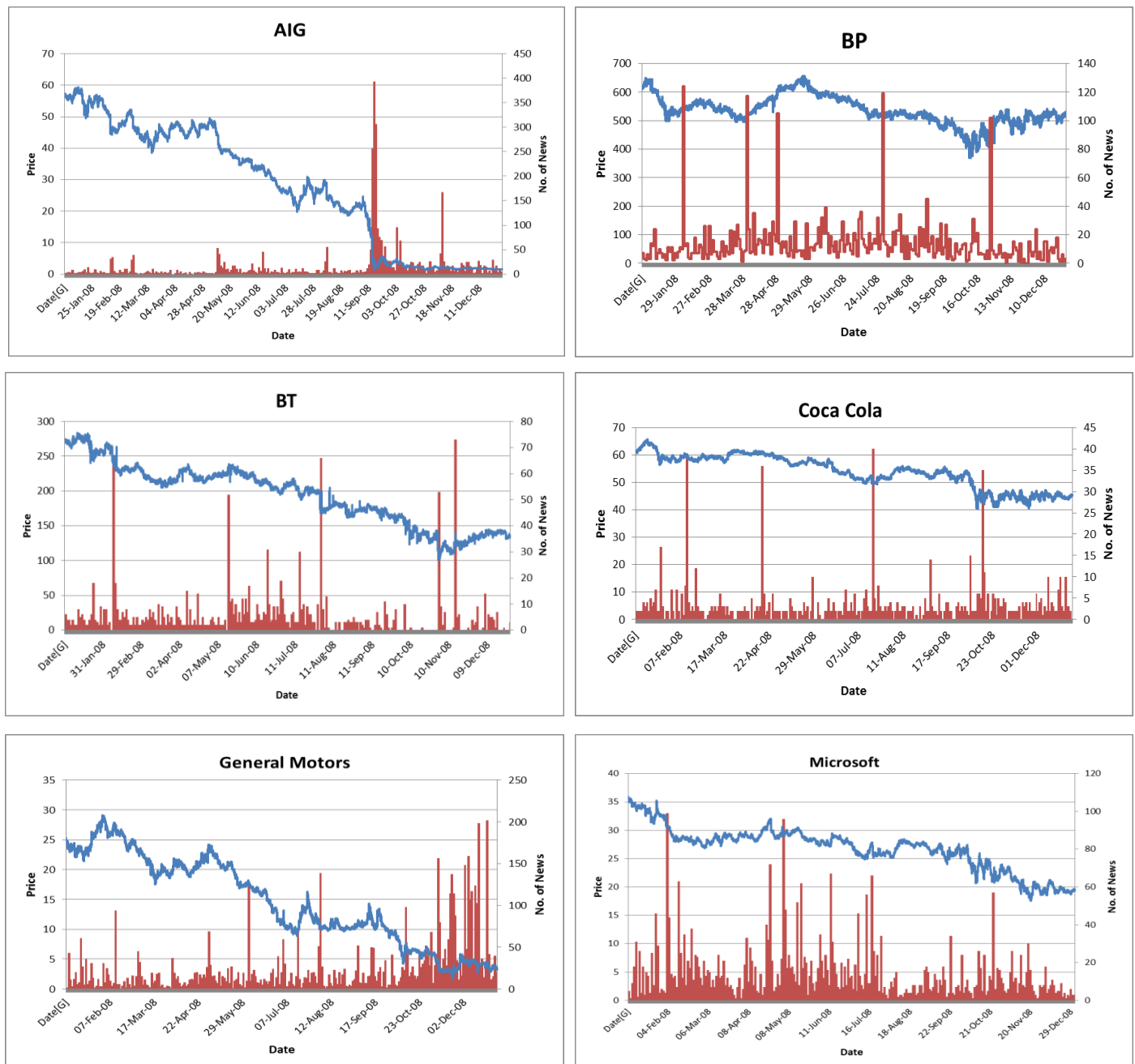


Figure 2.3: Closes prices (blue) plotted with number of news (red) for AIG, BP, BT Group, Coca Cola, General Motors and Microsoft in the year 2008.

In Figure 2.4, we display the logarithmic return against the logarithmic number of news; a negative relationship is clearly evident. That is to say, a company with a large amount of news will have lower average annualised return than those with less news. It is not surprising to observe that these stocks do display this trend in the period 2008; this was a period of global financial turmoil.

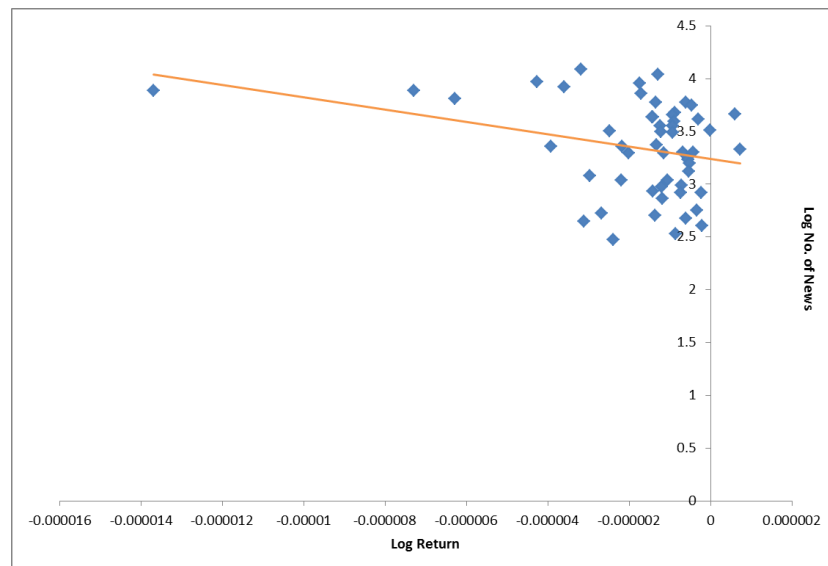


Figure 2.4: Graph showing the relationship between log return and log number of news.

The final preparation task for the data is to align the frequency of sentiment scores to the trading hours of 08:00-18:30. Any news item released before or after these trading times were summed and bucketed in the next time period, which may be the following day. As a consequence, there is no discarding of news sentiment which could be influencing the price and return thus, not a single piece of news data is ignored.

Company Name	Sector	Market Cap. (\$billions)	No. of News	Relevant News	Sentiment
AIG	Banking	81.36	7648	3870	-69.47
American Express	Banking	54.04	2261	1110	-9.90
AT&T	Telecommunication	207.08	4338	2430	16.53
Bank of America	Banking	475.94	9361	3283	-59.31
Chevron	Oil and Gas	161.51	3936	1519	-15.98
Coca Cola	Retail	262.55	1981	860	1.25
Disney	Media	54.1	2331	1080	3.76
Exxon Mobil	Oil and Gas	381.37	5900	2219	-21.80
General Electric	IT & Techonology	0.37	7680	3738	10.01
General Motors	Manufacturing	8.8	12176	6282	-102.32
Hewlett-Packard	IT & Techonology	84.40	3115	1536	4.24
Home Depot	Retail	44.73	1574	767	-6.79
IBM	IT & Techonology	118.89	3537	1911	20.77
Johnson & Johnson	Pharmaceuticals	177.42	1994	780	-3.51
JP Morgan	Banking	179.65	10971	4808	-5.45
Merck	Pharmaceuticals	139.22	3187	1535	-16.69
Microsoft	IT & Techonology	272.21	7245	3615	29.70
Pfizer	Pharmaceuticals	165.62	3091	1639	-5.47
Procter & Gamble	Retail	179.25	1995	887	3.48
Travelers	Insurance	18.11	972	612	-0.43
Verizon	Telecommunication	111.05	4536	3183	45.14
Wal-Mart	Retail	166.42	4633	1663	-13.85
Admiral Group	Insurance	3.86	474	344	1.30
ARM Holdings	IT & Techonology	1.97	1089	963	3.71
AstraZeneca	Pharmaceuticals	54.22	2135	1286	-6.27
Aviva	Insurance	22.01	5916	4800	22.47
Barclays	Banking	38.75	8987	5594	-47.25
BG Group	Oil and Gas	60.31	1700	1124	9.68
BP Group	Oil and Gas	161.80	5536	3436	-7.13
BskyB	Media	13.71	832	381	1.04
BT Group	Telecommunication	23.46	1957	1325	5.02
Burberry	Retail	2.73	525	410	0.23
GKN	Manufacturing	2.43	443	332	1.12
GlaxoSmithKline	Pharmaceuticals	99.64	3259	1869	-0.53
ITV	Media	3.14	1092	678	-4.21
Kingfisher	Retail	4.82	827	545	-1.58
Legal & General	Insurance	10.15	4292	3623	6.01
Lloyds Banking	Banking	28.70	8282	5476	29.00
Next	Retail	3.58	943	708	-5.33
Old Mutual	Insurance	8.06	1205	1057	2.59
Petrofac	Oil and Gas	3.03	503	418	2.24
Rolls Royce	Manufacturing	11.81	860	393	2.81
Royal Bank of Scotland	Banking	48.03	6409	2629	-1.82
Royal Dutch Shell	Oil and Gas	185.09	4137	2184	-15.55
RSA Insurance	Insurance	7.19	403	318	1.48
Sage Group	IT & Techonology	4.06	340	274	0.17
Shire	Pharmaceuticals	8.09	563	380	2.37
Standard Chartered	Banking	33.54	2289	1333	-0.10
Unilever	Retail	32.32	1317	1015	8.42
Vodafone	Telecommunication	122.37	4712	1788	15.71
Whitbread	Retail	3.14	730	625	5.72
Wood Group	Oil and Gas	3.11	295	223	0.76
WPP	Media	9.79	3572	2113	18.00

Table 2.5: Summary table of descriptive data for chosen assets within the period of 02/01/2008 – 31/12/2008.

3 Measures: Return, Volatility, Liquidity, Sentiment and Impact

3.1 Return, Volatility and Liquidity Measures

With the raw market data that is provided to us, we are able to calculate asset behaviour measures using close prices, bid prices and ask prices.

Bucket Size = 1 minute; Data Frequency = Minute Bar.

The **trading day** starts at 08:00 hours and ends at 18:30 hours thus in a trading day the total number of buckets is 630.

Any news and sentiment retrieved overnight (between 18:31 – 07:59 the following day) is put in the first time bucket of the following day i.e. 08:00. The same applies to weekend news, which is aggregated over both days and considered in the first minute when trading is resumed. This method of categorisation reflects our belief that reactions to news are reflected in stock price movements (or even days) after release.

Return

The return measure that we use in the model is the log-return calculated by the following equation

$$\text{Log}(R_t) = \text{Log}(P_t) - \text{Log}(P_{t-1}) \quad [3.1]$$

where P_t is the close price at time t . For convenience, $\text{Log}(R_t)$ are denoted by R_t throughout the rest of this paper.

Volatility

The volatility measure used in the model is calculated as a rolling standard deviation of log returns for one **trading day** (see section 2.3) leading to 630 data points.

Liquidity

Liquidity is represented by the spread of the bid and ask prices, which measures the cost one has to bear for immediate trade. Equation 3.2 gives the expression for bid-ask spread.

$$\text{Spread} = \frac{\text{Ask} - \text{Bid}}{\text{Mid}} \quad [3.2]$$

where Mid is the mid-price between the bid and ask prices.

3.2 Sentiment Measure

In our analytical model we have introduced two concepts, namely, (i) sentiment score and (ii) impact score. The sentiment score is a quantification of the mood (of a typical investor) in respect of a news event. The impact score takes into consideration the decay of the sentiment of one or more news events and how after aggregation these impact the asset price behaviour.

Sentiment Score

Thomson Reuters' news sentiment engine analyses and processes each news story that arrives as a machine readable text. Through text analysis and other classification schemes the engine then computes for each news event: (i) relevance, (ii) entity recognition and (iii) sentiment probabilities as well as a few other attributes (see Mitra and Mitra, 2011). As explained in Section 2.2, a news event sentiment can be positive, neutral and negative and the classifier assigns probabilities such that

$$Prob(positive) + Prob(neutral) + Prob(negative) = 1.0 \quad [3.3]$$

We turn these three probabilities into a single sentiment score in the range 0-100 using the following equation:

$$\hat{S}Sent = 100 * (Prob(positive) + \frac{1}{2}Prob(neutral)) \quad [3.4]$$

where $\hat{S}Sent$ denotes a single transformed sentiment score. We find that such a derived single score provides a relatively better interpretation of the mood of the news item. Thus the news sentiment score is a relative number which describes the degree of positivity and negativity in a piece of news. We further shift the sentiment score by subtracting 50 (the value corresponding to the neutral sentiment score = 0) to compute

$$SENT = \hat{S}Sent - 50 \quad [3.5]$$

Thus the score $SENT$ lies between -50 and +50. During the trading day, as news arrives it is given a sentiment value.

3.3 Impact Score

It is well known from research studies that newsflow affects asset behaviour (Patton and Verardo, 2012, Mitra, Mitra and diBartolomeo, 2009). Therefore, accumulation of news items as they arrive is important. Patton and Verardo (2012) noticed decay in the impact of news on asset prices and their betas on a daily timescale and further determine the complete disappearance of news effects within 2-5 days. It was also observed in Erlwein et al. (2013). Mitra, Mitra and diBartolomeo (2009) created a composite sentiment score in their volatility models after initial experiments saw no effects to volatility predictions with sentiment alone. Their decay period was over 7 days.

In order to compute the impact of news events over time we first find an expression which describes the attenuation of the news sentiment score. The impact of a news item does not solely have an effect on the markets at the time of release but also over finite periods of time that follow. To account for this prolonged impact, we have applied an attenuation technique to reflect the instantaneous impact of news releases and the decay of this impact over a subsequent period of time. The technique combines exponential decay and accumulation of the sentiment score over the time buckets under observation. We take into consideration the attenuation of positive sentiment to the neutral value and the rise of negative sentiment also to the neutral value and accumulate (sum) these sentiment scores separately. By separating the positive and negative sentiment scores we avoid an exact cancellation which may be interpreted as no news.

Data Granularity:

As explained in Section 2 the market data is taken as consolidated minute-bar data, that is,

Bucket Size = 1 minute; Data Frequency = Minute Bar.

The **trading day** starts at 08:00 hours and ends at 18:30 hours thus in a trading day the total number of buckets is 630.

Any news arriving overnight or during the weekend is bucketed into the next morning or days first minute. Hence, taking the assumption that the impact of such overnight and weekend news is incorporated into prices the following day.

Although news arrives asynchronously we work out the aggregated impact of all news in the following way.

Let

POS denote the set of news with positive sentiment value $SENT > 0$;

NEG denote the set of news with negative sentiment value $SENT < 0$;

and

$PNews(k, t_k)$ denotes the sentiment value of the k^{th} positive news arriving at time bucket t_k , $1 \leq t_k \leq 630$ and $k \in POS$; $PNews(k, t_k) > 0$.

$NNews(k, t_k)$ denotes the sentiment value of the k^{th} negative news arriving at time bucket t_k , $1 \leq t_k \leq 630$ and $k \in NEG$; $NNews(k, t_k) < 0$.

Let λ denote the exponent which determines the decay rate². The cumulated positive and negative sentiment scores for one day are given by equations 3.6 and 3.7.

$$PImpact(t) = \sum_{\substack{k \in POS \\ t_k \leq t}} PNews(k, t_k) e^{-\lambda(t-1)}, \quad t=1, \dots, 630 \quad [3.6]$$

$$NImpact(t) = \sum_{\substack{k \in NEG \\ t_k \leq t}} NNews(k, t_k) e^{-\lambda(t-1)}, \quad t=1, \dots, 630 \quad [3.7]$$

The arrival of more news items lead to higher values of accumulation; this therefore takes into account the news intensity, that is, the news flow. The impact is illustrated in Figure 3.1 which shows the impact score (with attenuation and accumulation) for the asset JP Morgan during the month of August 2008, with positive and negative sentiment represented separately.

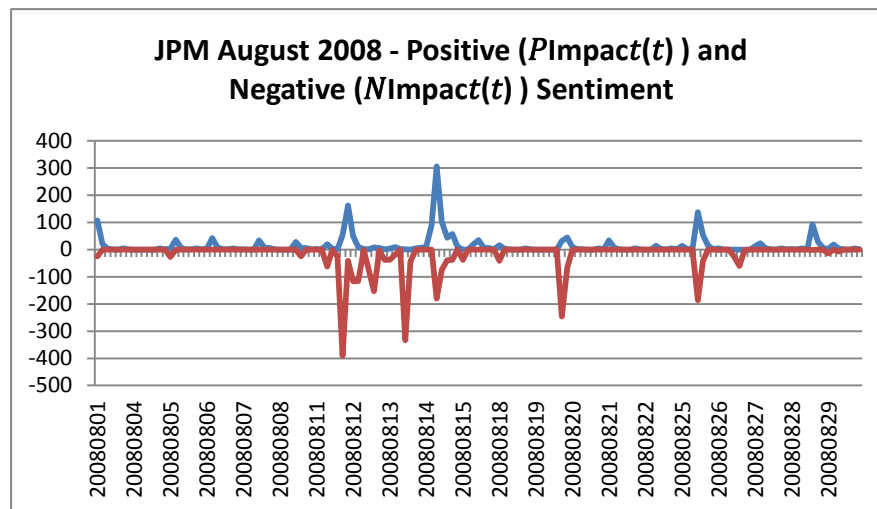


Figure 3.1: JP Morgan August 2008: News impact score (accumulated and aggregated) for positive (blue line) and negative (red line) sentiment respectively.

In order to compute the aggregated impact of all the news items which arrive during the day, we sum the positive and negative sentiments in each time bucket for the duration of the entire trading day. This gives us the impact score as set out in equation [3.8].

$$Impact_t = \sum_{i=1}^t PImpact(t) + NImpact(t), \quad t=1, \dots, 630 \quad [3.8]$$

For convenience, we express $Impact(t)$ also as $Impact_t$.

² We have chosen λ such that the sentiment value decays to half the initial value in a 90 minute time span. See Appendix A for our justification/heuristic of this choice.

4 The Model

Predictive analytical models for stock price return and volatility, such as the Autoregressive Conditional Heteroskedasticity ARCH model (Engle, 1982) and Generalised Autoregressive Conditional Heteroskedasticity GARCH model (Bollerslev, 1986), are well understood and applied extensively. These models exploit the techniques of time series analysis (Chatfield, 2009; Harvey, 1990) and are able to account for the correlation between each of the individual points in the time series of stock price returns by fitting them to models which are of Autoregressive (AR) nature. We have adopted this approach in the construction of our models for predicting return, volatility, and bid-ask spread. However, our method differs from ARCH/GARCH type models as it deals with two time series, (i) the first time series is that of market data and (ii) the second time series is the news metadata, see Figure 1.1. This approach can be seen as an extension of GARCH and AR models with the variable of news sentiment being the innovative addition. Another feature that distinguishes this from existing predictive models is the joint modelling of three key variables which describe asset behaviour, namely, return, volatility and liquidity, whereas the majority of existing literature focuses only on one variable, either return or volatility (Kothari, Zimmerman 1995; French, Schwert, Stambaugh 1987). We believe that apart from the classic measures of return and volatility that needs to be evaluated and monitored for risk control and trading strategies, it is also now imperative to include a liquidity measure as this reflects the conditions of the markets and feasibility of trades. This can be determined from two perspectives: (i) the spread of the market, evaluated by the bid-ask spread, and (ii) the depth of the market, computed by the total volume of bids and asks. We set out below a detailed description of the model:

Asset Return

Taking into consideration the inclusion of two time series, we construct an AR(2) predictive model for return with the enhancement of a news impact score (see section 4.2).

Let, R_t be the log return of stock prices at time period t
 R_{t-i} be the log return of stock prices by lag i
 $Impact_{t-1}$ be the news impact score of the previous time interval
 θ_i be the weighting coefficients to be estimated
 e_t be the error term

Therefore, the log-return R_t is given by the expression,

$$R_t = \theta_0 + \theta_1 R_{t-1} + \theta_2 R_{t-2} + \theta_3 Impact_{t-1} + e_t \quad [4.1]$$

Asset Volatility

The prediction model for volatility is an extended GARCH(1,1) model, as before with the addition of a news impact score.

Let, σ_t^2 be the volatility at time t
 ϵ_{t-1}^2 be the lagged log-return residuals

σ_{t-1}^2 be the lagged volatility
 $Impact_{t-1}$ be the news impact score of the previous time interval
 α_o be the weighting coefficients to be estimated
 u_t be the error term

Therefore, the volatility σ_t^2 is given by the expression,

$$\sigma_t^2 = \alpha_o + \alpha_1 \sigma_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \omega_1 Impact_{t-1} + u_t \quad [4.2]$$

Asset Liquidity

We measure the liquidity of the chosen asset by the bid-ask spread by constructing an AR(3) model with the addition of the news impact score.

Let, S_t be the bid-ask spread at time t
 S_{t-i} be the bid-ask spreads at lag i
 $Impact_{t-1}$ be the news impact score of the previous time interval
 γ_i be the weighting coefficients to be estimated
 η_t be the error term

The model for S_t is as follows:

$$S_t = \gamma_o + \gamma_1 S_{t-1} + \gamma_2 S_{t-2} + \gamma_3 S_{t-3} + \gamma_4 Impact_{t-1} + \eta_t \quad [4.3]$$

5 Computational Results and Validation

5.1 In Sample

The estimates for each weighting coefficient of the predictive models for the univariate case of all 53 assets are presented in Figure 5.1, Figure 5.2 and Figure 5.3 for the return, spread and volatility models respectively. The estimated parameters are displayed according to the colour scheme detailed in Table 5.1. From the fitting of the models it can be seen that news sentiment does indeed have an effect on the prediction of return, volatility and bid-ask spread as all coefficient estimates are non-zero.

<i>Model</i>	<i>Parameter</i>	<i>Colour</i>
Return	θ_0 – Intercept	Dark Blue
	θ_1 – 1 st lag of return	Green
	θ_2 – 2 nd lag of return	Red
	θ_3 – 1 st lag of Impact score	Purple
Volatility	α_0 – Intercept	Orange
	α_1 – 1 st lag of return residuals	Gold
	β_1 – 1 st lag of volatility	Turquoise
	ω_1 – 1 st lag of Impact score	Violet
Liquidity	γ_0 – Intercept	Light Blue
	γ_1 – 1 st lag of spread	Pink
	γ_2 – 2 nd lag of spread	Yellow
	γ_3 – 3 rd lag of spread	Light Green
	γ_4 – 1 st lag of Impact score	Grey

Table 5.1: Colour scheme applied in Figures 5.1, 5.2 and 5.3 to represent the estimated parameters.

Furthermore, our statistical hypothesis tests highlight the significance of the news variable to the predictive models. In particular, the following test was set up to investigate the significance of news effect on the volatility model:

$$H_0: \omega_1 = 0 \text{ vs } H_1: \omega_1 \neq 0.$$

This deemed to be highly significant at the 5% significance level for all assets (p-value = $<2e^{-16}$ for the volatility model) and so the null hypothesis is rejected, concluding that the estimator for the coefficient of news sentiment is not equal to zero and therefore has an impact on the prediction of volatility.

All intercepts of return are estimated to be negative apart from a handful of assets. The results of Microsoft showed to be the most extreme with values far greater than the average estimate of $-1.75e^{-6}$. In fact, the performance of Microsoft consistently stood out in the regressions of all three models.

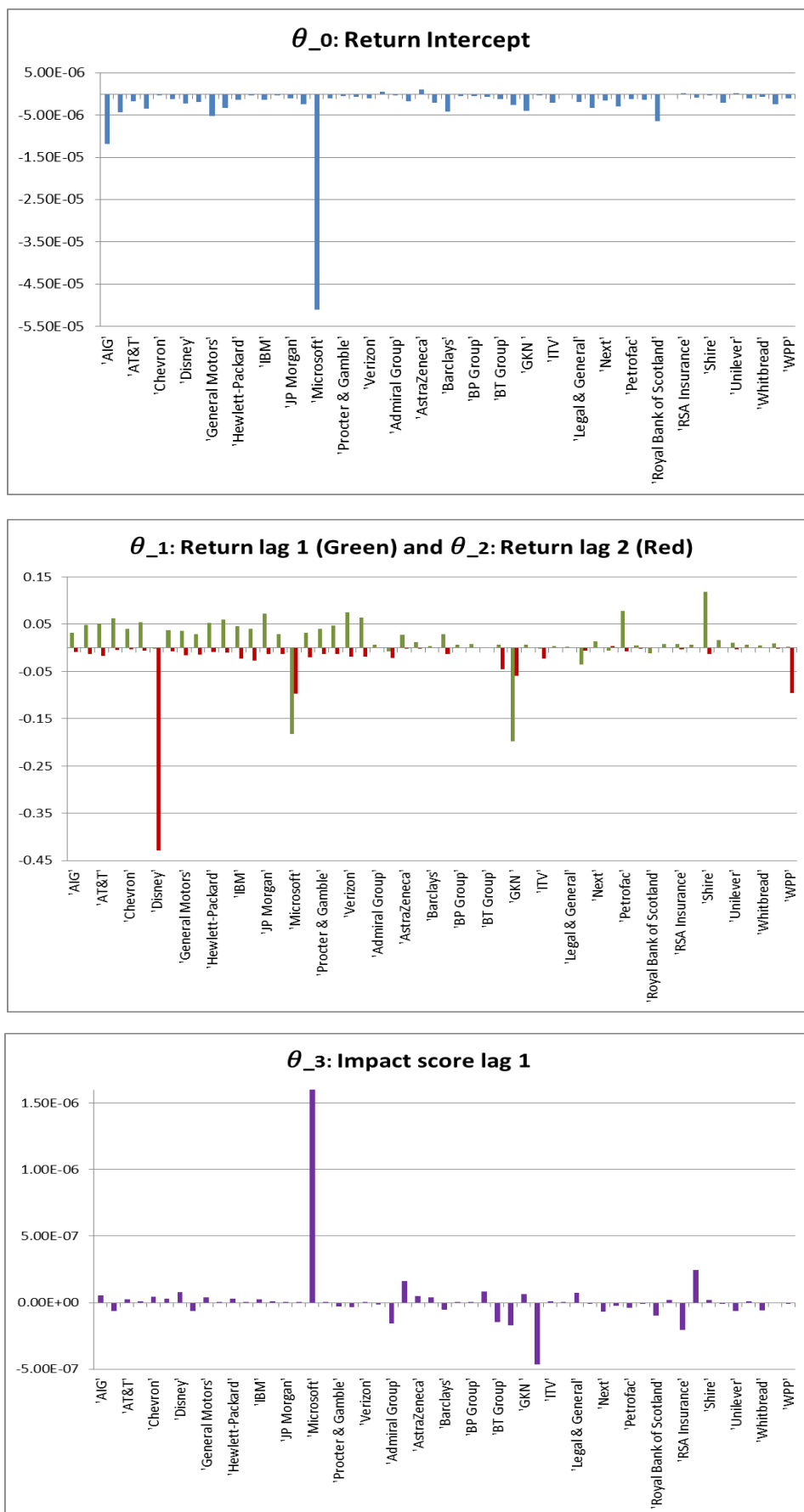


Figure 5.1: Distribution of 53 assets' estimated weighting coefficients for the lagged returns and impact score variable in the news enhanced return model.

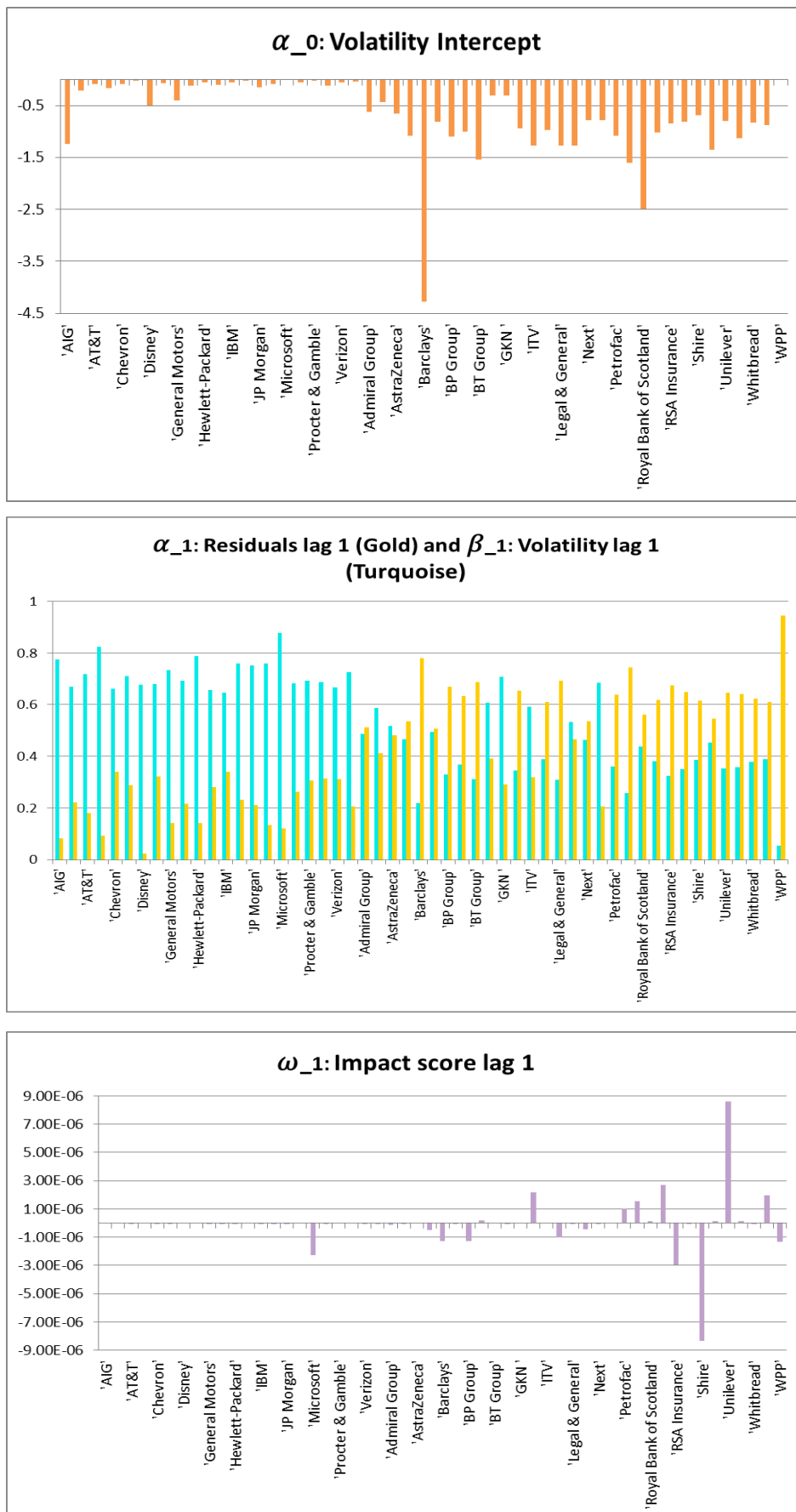


Figure 5.2: Distribution of 53 assets' estimated weighting coefficients for the lagged spread and impact score variables in the news enhanced spread model.

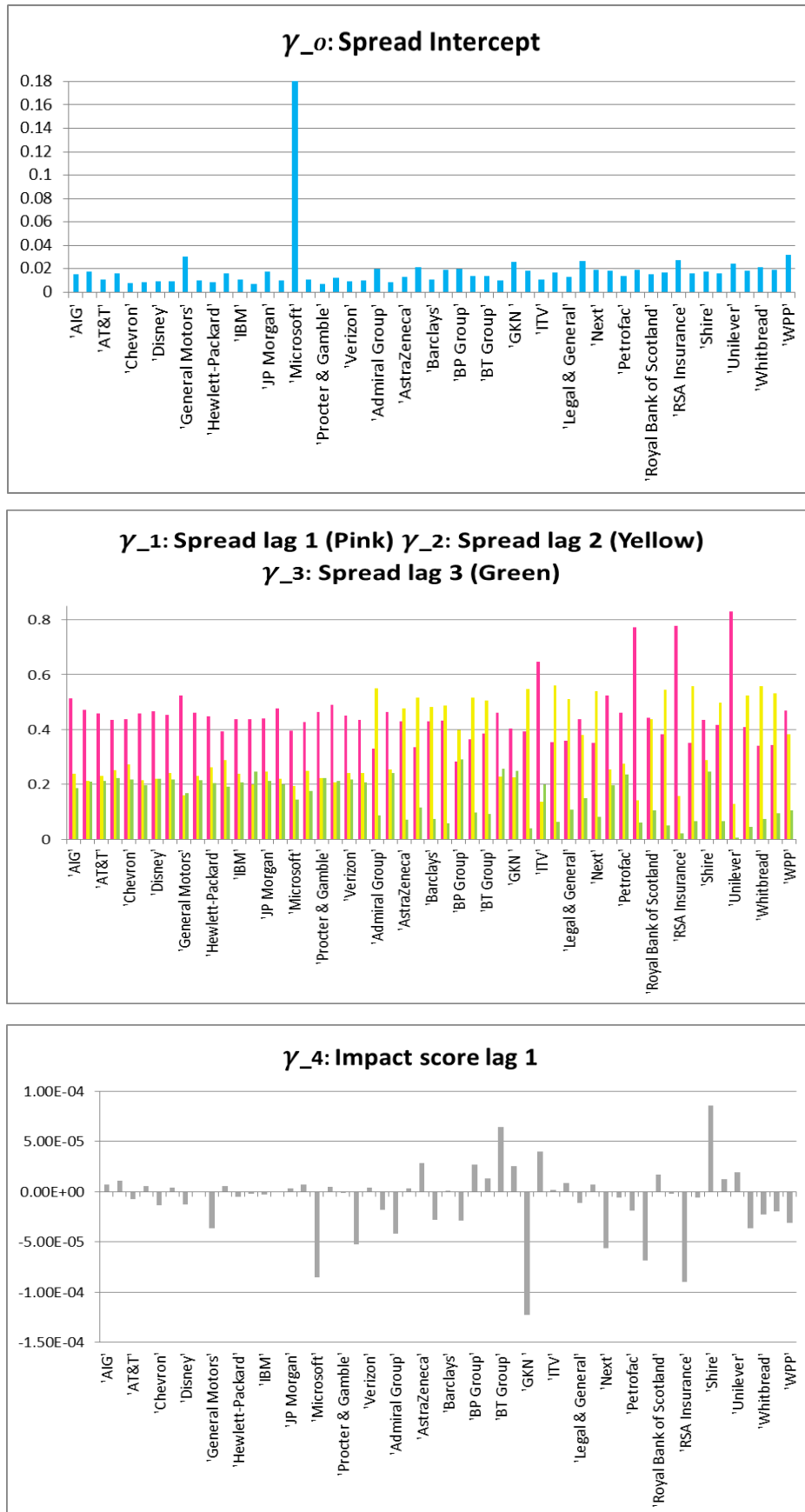


Figure 5.3: Distribution of 53 assets' estimated weighting coefficients for the GARCH and impact score variables in the news enhanced volatility model.

5.2 Out-of-Sample

To understand the extent to which news impact enhances the results of these models, we use the fitted model to predict out-of-sample values. To analyse the performance of these news enhanced models, we take the benchmark to be predictive models that only consider market data i.e. those described in section 4.1 without the exogenous variable of news impact, which would simply be AR and GARCH models. Direct comparison is made between the two sets of models to discover whether the news enhanced models perform better and hence whether news impact has a positive influence on the prediction of return, liquidity and volatility. The out-of-sample period is taken as January – March 2009. The conclusions interpreted from out-of-sample results are that news enhanced predictive models do better predict future values of liquidity and volatility but not for stock returns. Furthermore, the inclusion of news sentiment to a predictive model, in the form of an impact score, does increase accuracy of predictions of liquidity and volatility.

The predictive power of the fitted models is represented as the difference between the residuals from the market only model and the market with news model, i.e. prediction error. Prediction performance of the return model produces inconclusive results as there is a variation of positive and negative residuals across the assets. The prediction in the direction of return, and therefore price, is accurate however the magnitude of the return can be erroneous. This applies to both the market only model and the news enhanced model with little difference between them, as can be seen from Figure 5.4 where errors range from $6e^{-6}$ to $-6e^{-6}$. However, there are instances where the added news sentiment impacts the prediction of return in a more positive manner than the benchmark model, for example the Banking sector (see Figure 5.7). Figure 5.4 plots the average difference in prediction errors for each asset from the chosen universe of assets.

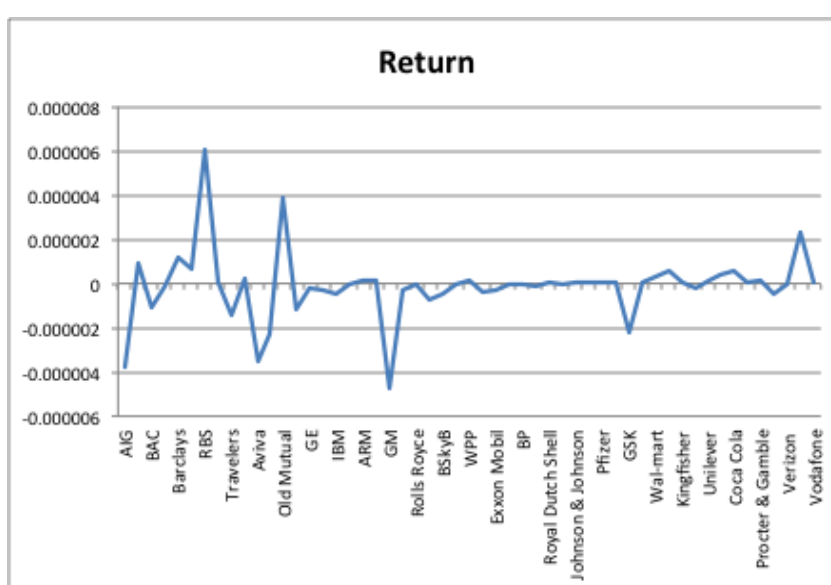


Figure 5.4: Average difference in prediction error between market only and news enhanced models for log-return for all assets in the period January-March 2009.

For the case of volatility, all assets performed better in the news enhanced GARCH model as indicated by positive differences in prediction errors throughout (see Figure 5.5). In other words, the residuals produced from the market only model were far greater than those from the news enhanced model, almost consistently by a factor of ten. Moreover, the plain market only model has a tendency to over predict the values of volatility. Therefore, the inclusion of news sentiment as an exogenous variable to the univariate GARCH model significantly reduces prediction errors and estimates values far closer to the true values of volatility. This is illustrated in Figure 5.5 which plots the difference in prediction errors for all assets.

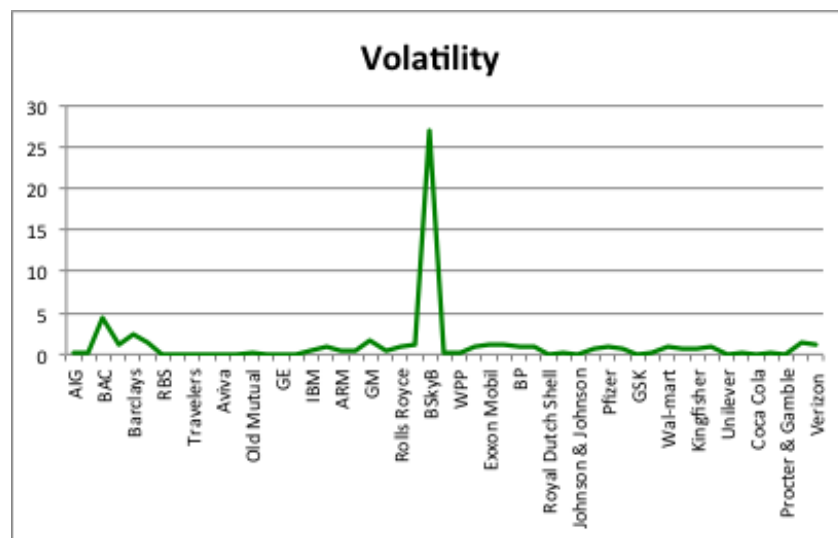


Figure 5.5: Average difference in prediction error between market only and news enhanced models for volatility for all assets in the period January-March 2009.

The difference in errors for liquidity (bid-ask spread) is relatively smaller than those seen for volatility with the majority of values falling within 3 decimal points (see Figure 5.6). It can be observed that positive values dominate in the plot of errors between the two models and an overestimation by the market data only model. Negative values do appear but only for major falls in spread. One explanation is that periods of low spread indicates a high volume of trading activity and such active trading periods are no longer instigated purely by the release of news. Rather more likely to be the influential factor for such instances are scheduled news announcements or market conditions, and hence a news enhanced model would be insufficient for predictions of such extreme negative values.

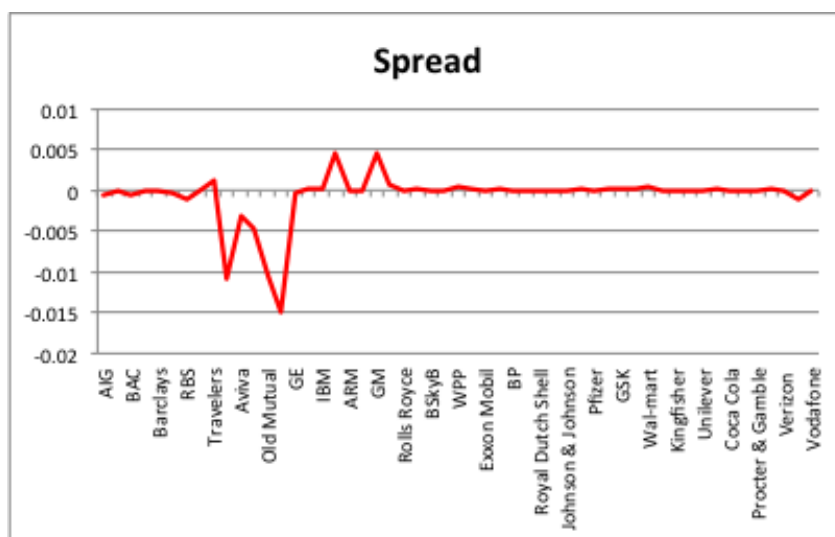


Figure 5.6: Average difference in prediction error between market only and news enhanced models for bid-ask spread for all assets in the period January-March 2009.

Figure 5.7 shows the performance of predictions for log-return, bid-ask spread and volatility by sector. Each value is the average difference in prediction error between the market only model and the news enhanced model across all assets within that sector. Not only does it clarify the superiority of prediction power by the news enhanced model for liquidity and volatility, but it also shows the best performing sectors within each measure. Banking and telecommunications ranked top for return predictions and manufacturing was worst. By contrast, the best performing sector for liquidity predictions was manufacturing as it had the highest positive error with a value of 0.0017 compared with insurance that had the most negative value of -0.0072. All sectors returned a positive error for volatility predictions, but the media industry distinctively outperformed the rest with an average error of 7.45, which is wildly distant from the true values of volatility that lie between 0-0.01.

Statistical measures such as root mean squared error (RMSE) and mean absolute error (MAE) were also calculated to compare prediction accuracy between the models. The results exhibit a very strong indication of superior performance in the prediction of volatility with the news enhanced model. Table 5.8 lists the rankings performance of each asset according to their return, spread and volatility prediction respectively. Through ranking the forecasting performance of each asset with respect to their RMSE, an obvious pattern emerges between return prediction and volatility prediction performance. From Table 5.8 it can be seen that there is a reversal in rankings for return and volatility, particularly in the highest and lowest ranks, with Microsoft showing the largest contrast in performance ranking last place for return prediction but first place for volatility prediction.

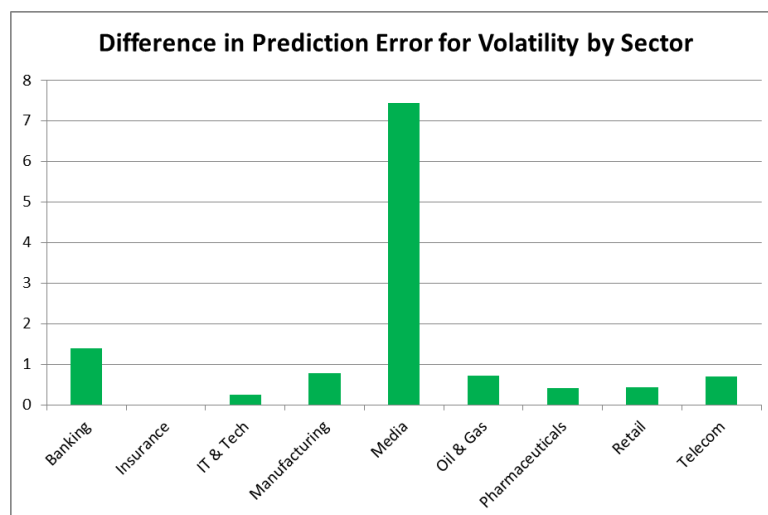
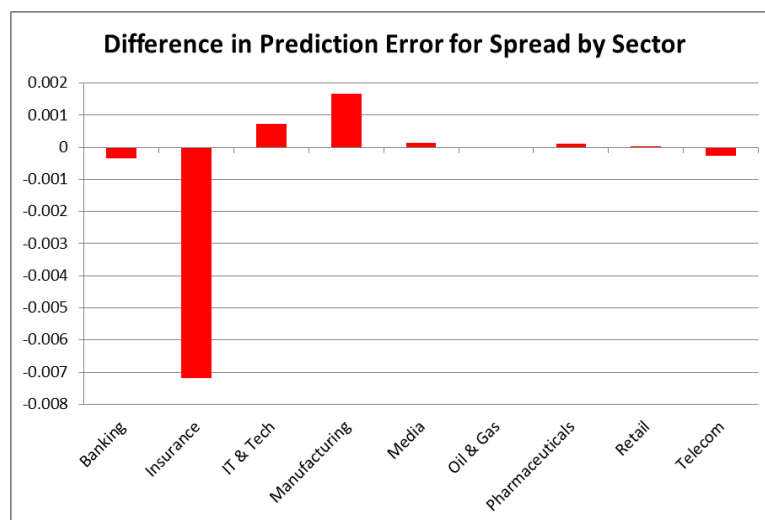
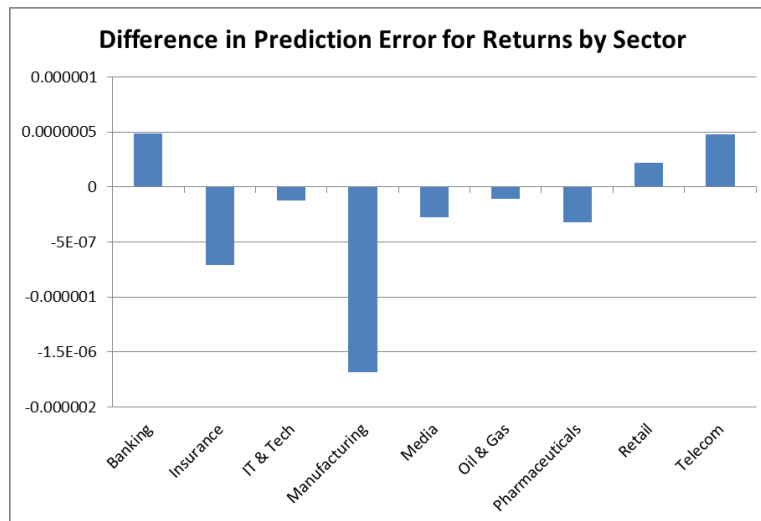


Figure 5.7: Average differences between out-of-sample prediction errors for market only and news enhanced model for log-return, bid-ask spread and volatility during the period January-March 2009. The figures are displayed by sector with the average value of each sector being composed of the mean residuals taken over 39060 points for all assets chosen within that sector.

Asset	Return Prediction Rank	Spread Prediction Rank	Volatility Prediction Rank
American Express	1	39	36
Aviva	2	29	15
Disney	3	9	31
Chevron	4	42	44
Hewlett-Packard	5	25	46
JP Morgan	6	53	39
Legal & General	7	11	9
General Electric	8	41	40
ARM Holdings	9	15	32
Bank of America	10	49	37
Burberry	11	46	35
The Home Depot	12	24	41
Travelers	13	36	38
Whitbread	14	38	22
ITV	15	14	8
Verizon	16	12	47
BT Group	17	45	6
Vodafone	18	1	12
Rolls Royce	19	30	5
Sage Group	20	21	24
Pfizer	21	26	48
BP Group	22	31	13
Royal Dutch Shell	23	22	16
Wood Group	24	20	20
Next	25	28	27
Kingfisher	26	37	18
BG Group	27	32	23
Standard Chartered	28	5	7
WPP	29	40	2
Coca Cola	30	8	51
GKN	31	2	34
Johnson & Johnson	32	23	53
Merck	33	10	43
Petrofac	34	13	14
Lloyds Banking	35	4	10
AstraZeneca	36	3	29
AT&T	37	33	42
Old Mutual	38	17	26
Wal-Mart	39	51	50
Shire	40	48	28
Admiral Group	41	34	30
BSkyB	42	7	17
IBM	43	19	49
Procter & Gamble	44	16	52
Unilever	45	35	25
RSA Insurance	46	47	21
General Motors	47	52	33
Royal Bank of Scotland	48	43	4
Barclays	49	27	3
Exxon Mobil	50	18	45
AIG	51	44	11
GlaxoSmithKline	52	6	19
Microsoft	53	50	1

Table 5.8: Forecasting performance ranking for each asset based on the root mean squared error for return, spread and volatility. Smallest errors receive the highest rank.

6 Future Research

We intend to extend this research by introducing models to study the behaviour of price, volatility and liquidity in a multivariate framework. We will consider correlations between companies and the relevant impact of news within companies from the same industry sector. We expect the prediction of returns to improve with such additional information, and in particular the prediction of liquidity and volatility. Hence we are planning to apply the multivariate formulations of AR and GARCH, that is, Vector Autoregressive (VAR) and Multivariate GARCH (MGARCH) models. Our argument for the multivariate approach is that it is more practical for trading strategies, which mainly apply several assets simultaneously to decide on trading positions. Moreover, we consider another avenue of investigation and view the relationship between the news impact score and our behavioural measures in an indirect manner such as Bayesian Inference. Thus the news impact score could be considered in the composition of the prior distribution which is then used to simulate the posterior distribution.

7 Discussion and Conclusions

Behavioural models and sentiment analysis are gaining momentum and acceptance within the investment community. Our study sets out to consider news sentiment and its impact on asset behaviour. We have introduced a novel concept of **news impact** which takes into consideration the volume of news and its decay over time. We are interested in predictive analysis of asset behaviour: return, volatility and liquidity so that this can be applied to construct trading strategies in an intraday setting. The findings are as follows. Enhancement of predictions of liquidity and volatility by inclusion of news meta data time series is claimed with supporting arguments for the representative data set. In particular the improvement of volatility predictions is substantial; this is in line with earlier studies as well as the reported results by other investigators. Liquidity prediction is also improved but not as much as that of volatility. Price prediction does not show any improvement at all and this is commensurate with results reported by other researchers. One explanation could be that market participants react quickly which leads to revision of price in response to news; however, volatility and liquidity lag this behaviour and hence is better predicted by the incorporation of news.

8 References

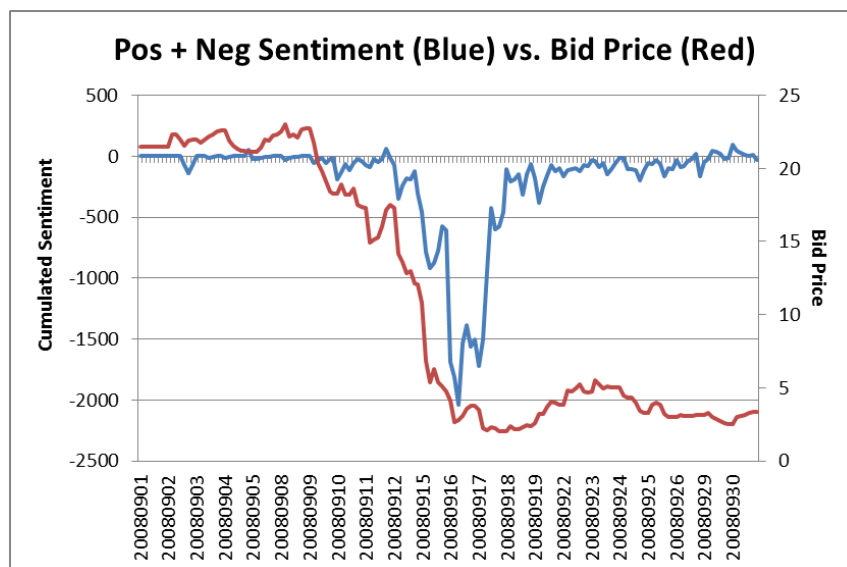
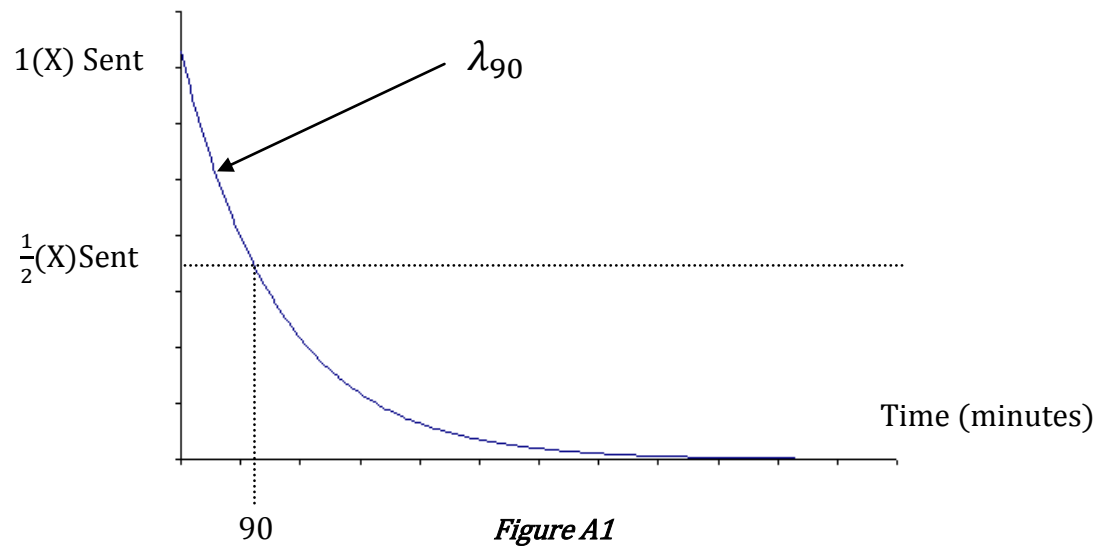
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9 Appendices

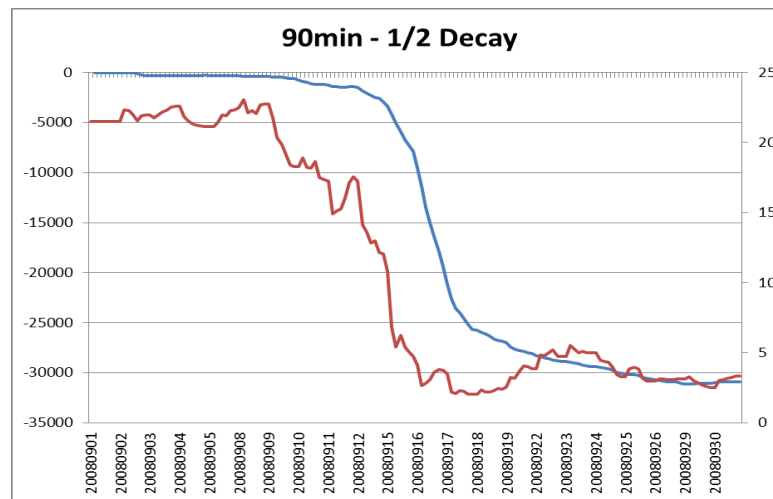
Appendix A



The sum of the sentiments for different arrivals of news events in a given time bucket of 1 minute duration (that is, 630 buckets during the trading day) gives the overall sentiment in that time bucket. In Figure A2 we plot the overall

sentiment for the asset AIG over one month of data in which the vertical axis on the left is the sentiment score, the vertical axis on the right is the bid price and the horizontal axis is $630 \times 21 = 13230$ minute-bar time buckets.

The work of Leinweber and Sisk (2011) and also our earlier work (Mitra, Mitra, and diBartolomeo, 2009) as well as the present study indicates that accumulation of the sentiment leads to a fairly good fit with asset price in general and with asset price volatility in particular.



The cumulated sentiment and asset prices are plotted and displayed in Figure A3.

Appendix B

Method and types of meta data:

IDN_TIME: The date and time of the news item (DD-MM-YYYY hh:mm:ss:sss). All messages are time-stamped to the nearest millisecond and is presented in GMT.

RELEVANCE: A real valued number from 0 - 1 indicating the relevance of the news item to the asset. It is calculated by comparing the relative number of occurrences of the asset with the number of occurrences of other organizations within the text of the item, with the exception to headline appearances where an immediate score of 1 is given if an asset is mentioned. The higher the relevance score is given, the more number of times an asset appears in the text.

SENTIMENT: This field indicates the predominant sentiment class for this news item with respect to this asset. The indicated class is the one with the highest probability and possible values are "1" (positive), "0" (neutral) and "-1" (negative).

SENT_POS: The probability that the sentiment of the news item was positive for the asset. The three probabilities sum to 1.

SENT_NEUT: The probability that the sentiment of the news item was neutral for the asset. The three probabilities sum to 1.

SENT_NEG: The probability that the sentiment of the news item was negative for the asset. The three probabilities sum to 1.